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An energy-efficient and cleaner production of hydrogen by steam reforming of glycerol using Aspen Plus

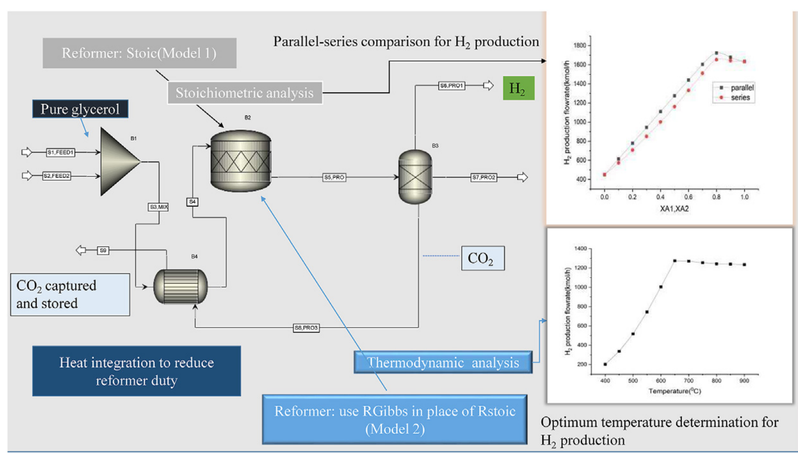
Md Alquma Haider, Nitin Dutt Chaturvedi *

Department of Chemical and Biochemical Engineering, Indian Institute of Technology Patna, Bihta, Patna, 801106 (Bihar) India

HIGHLIGHTS

- To develop an energy-efficient H_2 production model by steam reforming of glycerol.
- Maximum H_2 production with minimum reformer heat duty is found by thermodynamic feasibility of stoichiometric data.
- Reformer duty reductions show substantial potential for improving glycerol steam reforming efficiency.
- The model also incorporates CO_2 capture, resulting in cleaner H_2 production.

GRAPHICAL ABSTRACT



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ABSTRACT

Hydrogen (H_2) is produced in an efficient and sustainable manner using steam reforming of glycerol in this work. Two models are used to provide anticipation about how efficiently H_2 can be produced: Model 1, a stoichiometric model, and Model 2, a thermodynamic model. Model 1 helps determine the water/glycerol molar input ratio and reaction pathway. The thermodynamic feasibility of Model 1 is examined using Model 2, which then determines the optimum conditions for maximal H_2 generation. Model 2 produced 1275 kmol/h of H_2 with an H_2 mole fraction of 0.49 at 658 °C and 1 atm of pressure. The reactor's net heat load is greatly reduced through heat integration in this study. The energy required is reduced by 29.5% as compared to when no heat integration is used. The process of separating and

* *Corresponding author.*
E-mail address: nitind@iitp.ac.in (N.D. Chaturvedi).
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eliminating CO_2 is accomplished by employing the principles of mass and energy balance in the Aspen Plus.

1. Introduction

In recent times, there has been a growing trend among various nations to declare their intentions to transition their transportation and energy infrastructures towards more environmentally friendly fuel sources [1]. The phenomenon commonly referred to as the "global energy transition" is rapidly gaining momentum. There is a growing global trend towards the adoption of cleaner fuels as viable substitutes for conventional fossil fuels [2]. The utilization of cleaner fuels is essential in the continuous effort to reduce carbon emissions, reduce global warming, and promote sustainable development [3]. Renewable energy sources, such as solar power, wind power, hydroelectric power, and geothermal energy, as well as biofuels and hydrogen (H_2), have emerged as prominent alternatives to fossil fuels and are experiencing increasing levels of public interest [4]. H_2 is considered a viable fuel option for various sectors, such as transportation and electricity generation, due to its clean and carbon-free characteristics [5]. In other words, H_2 is regarded as a promising fuel due to its versatile applications, including its potential as an energy source, energy storage medium, fuel for combustion, and as a means to mitigate carbon emissions [6]. H_2 is often considered to be the most promising energy carrier, especially as a fuel for fuel cell technologies [7]. Additionally, H_2 finds extensive utilization in industrial processes, including but not limited to petroleum refining [8], ammonia production [9], methanol synthesis [10], and steel manufacturing [11]. Researchers are currently working in efforts to enhance the performance characteristics of the subject, which encompass a high energy density, significant capacity, extended lifespan, and convenient storage and transmission capabilities [12,13]. H_2 can be produced using various methods, such as electrochemical, thermochemical, photochemical, photocatalytic, and photoelectrochemical processes [14]. Coal and natural gas, along with alternative sources such as solar, nuclear, and biomass, have the potential to serve as viable feedstocks for the production of H_2 [15]. In order to effectively mitigate environmental impacts, it is essential to ensure the eco-friendly production of H_2, particularly through the utilization of renewable feedstocks [16]. The rapid expansion of the biodiesel industry has resulted in a significant increase in the production of crude glycerol [17,18]. Although the cosmetics, food, and pharmaceutical industries utilize purified glycerol, there is too much of it on the market for biodiesel producers to use, and the cost of purification is considerable [19,20]. One potential approach involves the utilization of crude glycerol as a precursor for the production of H_2 [21,22]. Pure glycerol is utilized by numerous researchers in conjunction with various catalysts such as Ni/CeO_2 [23], Ni/Al_2O_3 [24], NiO/$NiAl_2O_4$ [25], Ni/Fly-ash [26], among others [27]. The production of biodiesel involves the utilization of a technique known as *trans*-esterification, which results in the formation of glycerol ($C_3H_8O_3$) as a byproduct. Glycerol is also called glycerine or 1,2,3-propane tri-ol [28].

Several ways to make H_2 have been thought of, including steam reforming (SR) [29], partial oxidation [30], dry reforming [31], water phase reforming [32], and autothermal reforming [33]. The SR process has been identified as a highly efficient technique for H_2 production, owing to its superior performance in this regard and its relatively low environmental impact when compared to alternative methodologies [34]. The SR method is basically a chemical change of hydrocarbons and steam into H_2 and carbon oxides. It has three main steps: reforming or making synthesis gas (syngas), water-gas shift (WGS), and methanation or cleaning the gas [35]. In the context of glycerol steam reforming (GSR), it has been postulated that the reaction between water vapor and glycerol yields primarily carbon monoxide (CO), carbon dioxide (CO_2), and H_2. The process described above is considered the most efficient pathway for conversion, as it leads to increased selectivity and yield of the final product by directly removing H_2 from water [36]. The topic of interest in recent years has been the valorisation of glycerol through the process of GSR [37]. The primary reaction occurring in the process involves the decomposition of glycerol, while the WGS reaction is a secondary reaction [38]. The aforementioned process occurs under conditions of elevated operating temperatures and atmospheric pressure, and is commonly referred to as an endothermic reaction [39]. The process of vaporizing reactants necessitates a substantial amount of energy, thereby reducing the overall energy efficiency of the process. The variables that influence the efficiency of H_2 production through the GSR process include temperature, pressure, water to glycerol feed ratio (WGFR), feed reactants to inert gas ratio, and feed gas rate [40]. In the last ten years, a lot of research has been done on GSR using different catalysts, such as transition metals (like Co and Ni) and noble metals (like Pt, Rh, and Ru) based on different oxides [41]. Ni-based catalysts have been studied a lot in the past few years [42]. They were found to be the most popular choice in the literature. Using a Ni–Cu–Al catalyst in a continuous flow fixed-bed reactor at 500–600 °C and normal pressure, Dou et al. (2014) found that 54.3–70.4% H_2 could be made from GSR [43]. Based on a study, sorption-enhanced GSR was found to be a good way to increase the quality of H_2 to more than 90% and reduce the amount of CO_2. The effect of temperatures between 400 and 700 °C on the selectivity of the product is strong. As the temperature goes up, the H_2 selectivity goes up [44]. A study by Buffoni et al., 2009 shows that nickel catalysts are active and selective, with a strong relationship between the reaction temperature and the gaseous products made from glycerol. This showed that the lowest temperature needed to get H_2 with a high sensitivity is 550 °C [45]. The findings of the study indicate that optimal conditions for H_2 production involve a temperature exceeding 900 K and a molar ratio of 9:1 between water and glycerol. In the given circumstances, the generation of methane is effectively suppressed, and the thermodynamic principles prevent the accumulation of carbon. According to the literature, the maximum amount of H_2 that can be produced per mole of glycerol is 6, whereas the maximum amount predicted by stoichiometry is 7 [46]. When the temperature is raised to 350 °C and a feed containing 10% weight glycerol is used, the Pt/SiO_2 catalyst exhibits a H_2 yield of 60% [47]. When employing a WGFR of 3:1 and operating within a temperature range of 500–600 °C, the utilization of $ZrO_2/Ni/Al_2O_3$ catalyst results in a yield of 70% [48]. At a temperature of 650 °C and a WGFR of 6:1, the resulting yield of H_2 is 65.64% [49].

The aforementioned studies conclude that GSR is an effective method for producing H_2. However, the production of steam is an energy-intensive process because GSR as a whole is endothermic. So, the heat load of the reactor has to be kept as low as possible. The decomposition of glycerol at higher temperatures also reduces the amount of H_2 that can be produced. It is clear that a GSR model is needed that directly addresses these two problems. The developed model offers a way to produce H_2 that uses less energy. The model is developed using Aspen Plus software, which incorporates carbon capture and storage (CCS) advancements. Here are some of the unique features and improvements that the proposed model offers.

- ➢ The energy required in this model is much less than in models available in the literature;
- ➢ The developed stoichiometry-based model can be applied to unknown kinetics;
- ➢ This work allows to give a relationship between conversion and temperature for H_2 production.
- ➢ This model also incorporates CCS, making it an environmentally friendly production model.

The rest of this work is put together in the following way: The procedure for creating the GSR process simulation model in Aspen Plus is outlined in Section 2. In Section 3, the model is initially validated by comparing it with previously published literature. Several effects on H_2 production have been reported. Model 1 first examines the impact of the water/glycerol molar feed ratio. The optimal value determines whether reactions are conducted in parallel or in series. Model 2 then verifies the feasibility of Model 1's results and establishes a relationship between the two models regarding conversion and temperature. Afterward, the reduction of reformer heat duty is compared to the literature based on possible results. Finally, Section 4 gives the conclusion of this study in detail.

2. Materials and methodology

The present section delineates the procedural steps entailed in formulating the GSR process simulation model within the Aspen Plus software. The process begins with the identification of the raw materials and their corresponding reactions. Two Aspen Plus simulation models, namely RStoic and RGibbs, serve as the basis for the model's development. Additionally, different input parameters are incorporated into the simulation models.

2.1. Materials used

Pure glycerol and water are the raw materials that are used to make H_2. Glycerol is a sticky liquid that is thick when it is cold, has no smell or colour, and tastes sweet. Because it has three hydroxyl groups, it can take water from the air because it is hygroscopic. Glycerol is slightly thicker than water. Three hydrophilic hydroxyl groups in the liquid glycerol are what give it its hygroscopic properties and water solubility [50].

H_2 production via GSR is a high-temperature process that converts glycerol to synthesis gas, a H_2-rich gaseous product. The chemical reaction (Eq. (1)) involved in the GSR into syngas is well understood and has already been described in the literature [51].

$$C_3H_8O_3 \leftrightarrow 3CO + 4H_2 \; (\Delta H_r^0 = +250 \; kJ/mol) \tag{1}$$

The overall reaction for GSR is represented by the following equation [52].

$$C_3H_8O_3 + 3H_20 \rightarrow 3CO_2 + 7H_2 \; (\Delta H_r^0 = +128 \; kJ/mol) \tag{2}$$

CO is converted to CO_2 as a result of an excess of vapor in the process, which also results in the release of more H_2 gas. Since the reaction is reversible, both species exist in dynamic chemical equilibrium. This reaction is known as the WGS [53]:

$$CO + H_2O \leftrightarrow CO_2 + H_2 \; (\Delta H_r^0 = -41 \; kJ/mol) \tag{3}$$

Since GSR reactions are conducted at extremely high temperatures, there is a significant chance of glycerol decomposition (Eq. (4)) into oxygenates with lower molecular weights and lighter hydrocarbons such as CH_4, C_2H_6, etc. [54].

$$C_3H_8O_3 \rightarrow C_xH_yO_z + Carbon + Gases(H_2, CO, CO_2, CH_4, C_xH_{x+2}) \tag{4}$$

Coke formation occurs in the system at high temperatures, which can result in depositions on the catalyst surface and subsequent inactivation. Carbon deposition is a very severe problem in heterogeneous catalyst-based catalytic SR systems [55].

2.2. Methodology and detailed process scheme

Fig. 1 presents the GSR process model developed by Aspen Plus based on stoichiometry. The simulation's numerous components include glycerol ($C_3H_8O_3$), water (H_2O), oxygen (O_2), hydrogen (H_2), carbon dioxide (CO_2),2-methoxyethanol ($C_3H_8O_2$), propane (C_3H_8), ethane (C_2H_6), methane (CH_4), carbon monoxide (CO), and carbon (C). These components are regarded as typical components with well-known chemical formulas. UNIFAC has been chosen as the property method [56]. Rather than relying on molecular contributions, UNIFAC (UNIversal Functional Activity Coefficient) is based on group

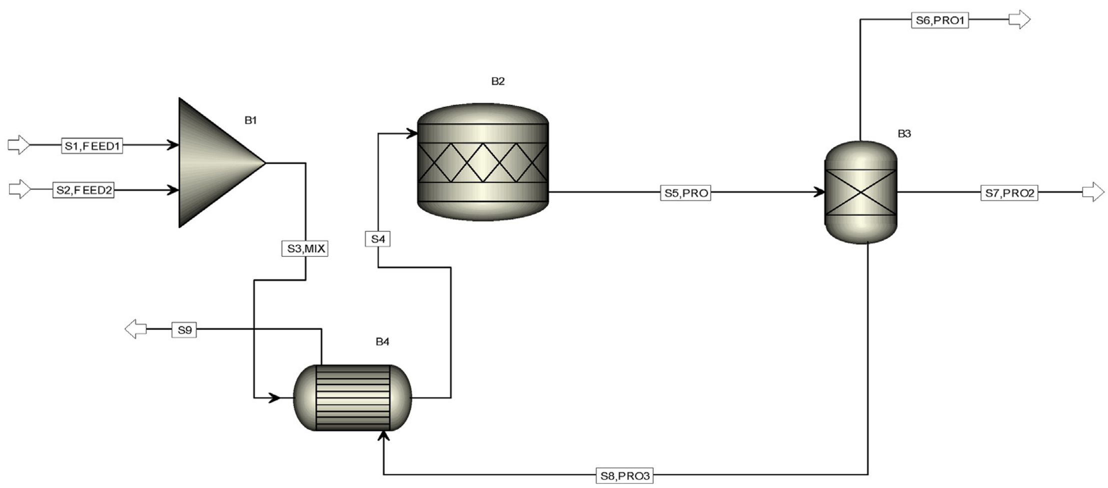

Fig. 1 – The process flowsheet for H_2 production based on stoichiometry (Model 1).

contributions. "UNIFAC" can forecast activity coefficients using a small set of group parameters and group-group interaction parameters. The "UNIFAC" model is highly predictive because it is a group-contribution model. At low pressure (below 10 atm), it is utilized to represent severely nonideal liquid mixtures [57]. In this procedure, two feed streams—pure glycerol and water—were fed into a mixer (MIX) before being fed to the reformer. Before entering the reformer, the mixed stream was heated up to a temperature calculated based on optimization of heat exchanger size using one of the product streams that is available at product temperature. At ambient temperatures (25 °C and 1 atm), glycerol and water were given in a specific molar ratio [56]. A block designated as a stoichiometric reactor (RStoic) was the steam reformer. RStoic can simulate parallel or series reactions. Additionally, RStoic has estimations for product selectivity and heat of reaction. Stoichiometry reactor models' calculations are based on equations for the material and energy balances [58]. The aforementioned GSR reactions occur at atmospheric pressure and temperatures between 400 and 750 °C and follow the stoichiometry specified in Eqs (2) and (4). The reformer temperature range was selected in accordance with research that was conducted for the GSR process [59]. The reactor's output is routed to Block B3, which allows for the assignment of flow rates or componential split fractions for each component in each of the (n - 1) product streams based on the inputs' combined feeds. B3 generates three streams: S6, PRO1; S7, PRO2; and S8, PRO3. S6, PRO1 is pure H_2, S8, PRO3 is CO_2, and S7, PRO2 is a combination of all other components. In order to lower the reformer's heat load, the S8, PRO3 product stream is being utilized to heat up the S3, MIX stream. When S4 is heated, it goes into B2 (Refer Table 1 for block description). To achieve enhanced energy efficiency and promote environmentally friendly production practices, it is imperative to implement measures for the capture of CO_2 and the minimization of reformer heat duty. The separation of CO_2 from S5, PRO can be achieved by employing the split fraction method. The subsequent equations are employed to perform the process of splitting fractions into products. The variable f_i^k denotes the quantity of component i present in feed stream k.

Table 1 – Description of blocks of the given flowsheet.

Blocks	Description
B1	Mixer (combines an arbitrary number of input streams and produces a single stream by simple material balance)
B2	Stoic reactor (used when the reaction stoichiometry is known but kinetics is unknown.)
B3	SEP (use to the assignment of the flow rates or componential split fraction of each component)
B4	HeatX (use to heat mixed stream in order to reduce reactor heat duty)

Consequently, the amount of component i that enters the block F_i can be determined by Eq. (5) [60].

$$F_i = \sum_{k=1}^{l} f_i^k \tag{5}$$

Eq. (6) provides the relationship between the split fraction α_i^j of a specific component i in stream j and the quantity of component i leaving the block in stream j, S_i^j [60].

$$S_i^j = \alpha_i^j F_i \tag{6}$$

The captured CO_2 is transferred to a heat exchange process with S3, MIX. This process aims to raise the temperature of S3, MIX to a maximum value determined through the sensitivity analysis of B4. When the temperature of S3, MIX increases, it is subsequently transferred to the reformer as S4. It should be noted that S5, PRO can be directed toward a CCS system, which can be accomplished through processes such as absorption/stripping, cryogenic separation, membrane separation, and pressure swing adsorption, among others. However, current research focuses on H_2 production, so SEP is currently used for CO_2 separation (Refer Table 2 for streams description).

Process Assumptions The following assumptions were considered in the development of the model.

Table 2 – Description of streams of the given flowsheet.

Streams	Description
S1, FEED1	Pure glycerol (25 °C and atmospheric pressure)
S2, FEED2	Water (25 °C and atmospheric pressure)
S3, MIX	Mixed stream (cold)
S4	Mixed stream (hot)
S5, PRO	Products from B2
S6, PRO1	H_2 (pure)
S7, PRO2	Mixed products from B2
S8, PRO3	CO_2 using as heating medium
S9	CO_2 after losing heat

- The process occurs under steady-state conditions.
- Pre-treatment of feed is not required since pure glycerol is taken as feed instead of crude glycerol.
- Negligible pressure drops in B1.
- Vapor phase is only valid in B2.
- Shortcut method is used in HeatX.

The results of Model 1 are based on stoichiometric analysis, which should be verified by thermodynamic analysis. In Model 2 use RGibbs in place of RStoic as B2. All other variables have the same notation as Model 1.

Several factors, like the amount of H_2 gas production, can affect how well the GSR process works. The GSR operating conditions chosen for the Aspen Plus simulation of the process model are listed in Table 3. For both model, molar feed ratio of water and glycerol is varied. For Model 1, the conversion of glycerol in Eqs (2) and (4) were varied to see how they affected the production of H_2. For example, if the conversion of glycerol in Eq. (2) is 90%, then the conversion in Eq. (4) is set to 10%. For Model 2, temperature of reformer is varied from 400 to 750 °C and then up to 900 °C for better analysis.

3. Results and discussions

This section presents a comprehensive analysis of the diverse factors influencing the production of H_2. Firstly, the model is validated by comparing it with other existing literature on H_2 production. The investigation of the impact of the molar feed ratio of water to glycerol is conducted in Model 1. The optimal value is employed to assess whether reactions are conducted in a parallel or series manner. The verification of the feasibility of the results obtained from Model 1 is conducted through the utilization of Model 2. This subsequent analysis establishes a relationship between the two models in relation to the variables of conversion and temperature. Subsequently, an analysis is conducted to compare the reduction of reformer heat duty achieved in this study with existing literature, yielding promising and viable results.

3.1. Model validation

Table 4 presents a comparative analysis of the mole fraction of H_2 values derived from GSR, lined up with relevant literature studies. The majority of GSR in the literature was done experimentally. In the few simulation studies that have been published, the impact of temperature is typically examined. Below is a summary of the values found in the literature.

It should be noted that Model 1 is based on the principles of stoichiometry, thus necessitating the verification of its results for thermodynamic feasibility. This verification process is carried out in Model 2.

3.2. Effect of water/glycerol molar feed ratio

The effect of the water-to-glycerol feed ratio on the rate of H_2 production is examined and depicted in Fig. 2. The entire feed rate is assumed to be 1000 kmol/h. The molar quantity ratio fluctuates between 0.11 and 9. As glycerol molar flowrate

Table 4 – Comparison of conversion values with those in the literature.

Reaction temperature (^{0}C)	H_2 fraction	Reference
400	0.65	[61]
727	0.45	[62]
700	0.60	[54]
600	0.28	[63]
750	0.26	[64]
658	0.49 (Model 2 reformer: RGibbs)	This study
N.A	0.66 (Model 1 reformer: RStoic)	This study

Table 3 – Model's equipment and input stream conditions (only manually entered variables were filled; values left for Aspen to calculate are denoted by a – symbol).

Blocks/streams	Temperature (^{0}C)	Pressure (atm)	Vapor fraction	Split fraction	Flowrate (kmol/hr)
S1	25	1			100–900
S2	25	1			900–100
B1	–	1			
B2	–	1	1		
B3	–	–		$H_2 = 1$ for S6,PRO1 $CO_2 = 1$ for S8,PRO3	
B4	–(outlet temperature of cold stream is calculated via optimization)	–			
S8,PRO3		1			

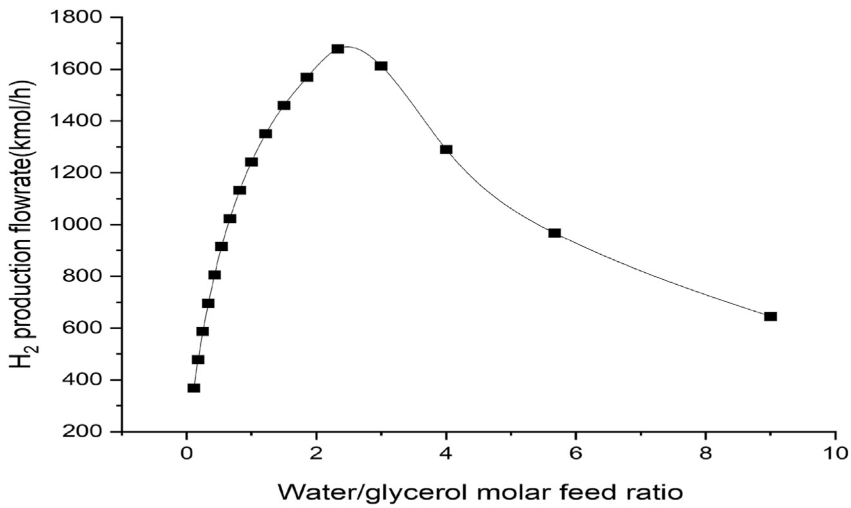

Fig. 2 – H_2 production flowrate vs. water/glycerol molar input ratio.

increases from 100 to 300 kmol/h and water molar flowrate decreases from 900 to 600 kmol/h, H_2 production increases and reaches a maximum, but further increases in glycerol flowrate and decreases in water flowrate result in a decrease in H_2 production. Reportedly, the optimal molar feed ratio is 2.33.

3.3. Effect of conversion of glycerol in parallel reaction

In a parallel reaction, many reactions take place concurrently, typically with the same reactants but distinct end products. How much time is spent on each pathway depends on the rates of the parallel reactions. The supply of reactants for slower reactions will decrease as faster reactions use more of them. The major reaction (R1) leading to the formation of H_2 is the GSR, which is Eq. (2). The decomposition of glycerol (R2) is represented by Eq. (4). Fig. 3 shows the results of running both reactions in parallel. As the value of XA1 increases from 0 to 1, there is a rapid increase in the production of H_2. Conversely, when XA2 rises from 0 to 1, the production of H_2 exhibits a slower rate of increase. If both R1 and R2 occur in parallel fashion, then H_2 production will first increase and then decrease. Table 5 displays H_2 production data for a few representative positions. Fig. 3 shows that R1 has a quicker reaction time compared to R2. This is because when XA1 = 1, indicating total conversion in R1, H_2 production increases.

3.4. Effect of conversion of glycerol in a series reaction

The occurrence of a series of reactions can result in the formation of a particular distribution of products. The final product is determined by both the order in which the reactions take place and the relative rates at which each step proceeds. The reaction conditions and kinetics of each step have an influence on the selectivity and yield of the desired products. The present study focuses on the modelling of product formation and distribution, with a particular emphasis on the conversion of reactions. The reformer utilized in Model 1 operates based on the fundamental principle of stoichiometry. Fig. 4 illustrates the results of the simulation

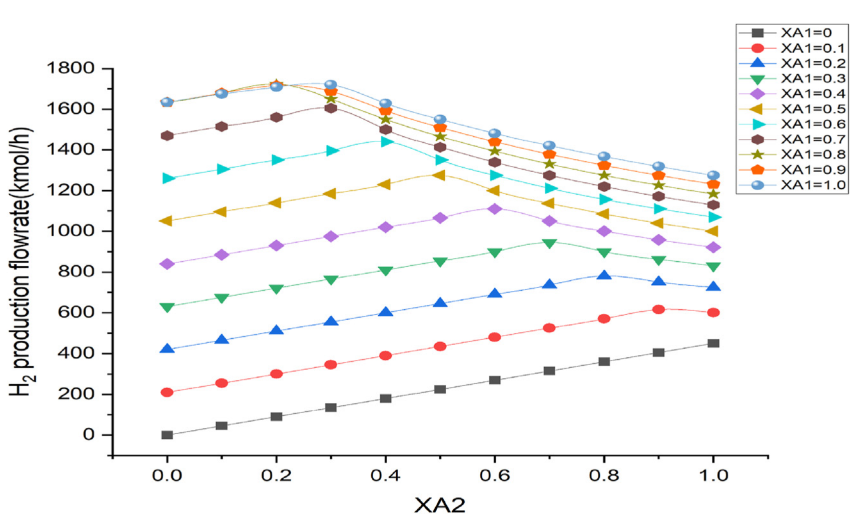

Fig. 3 – Effect of conversion of glycerol in parallel reaction for Model 1.

Table 5 – H_2 production flowrate for both reactions when running in parallel for some selected conversion.

XA1	XA2	H_2 production (kmol/h)
0	1	450
0.1	0.9	615
0.2	0.8	780
0.3	0.7	945
0.4	0.6	1110
0.5	0.5	1275
0.6	0.4	1440
0.7	0.3	1605
0.8	0.2	1723.33
0.9	0.1	1678.33
1	0	1633.33

Continuously as XA2 ranges fro m 0 to 1, and when XA1 = 0.9, indicating incomplete conversion, H_2 production increases first, reaches a maximum at XA2 = 0.9, and then drops further.

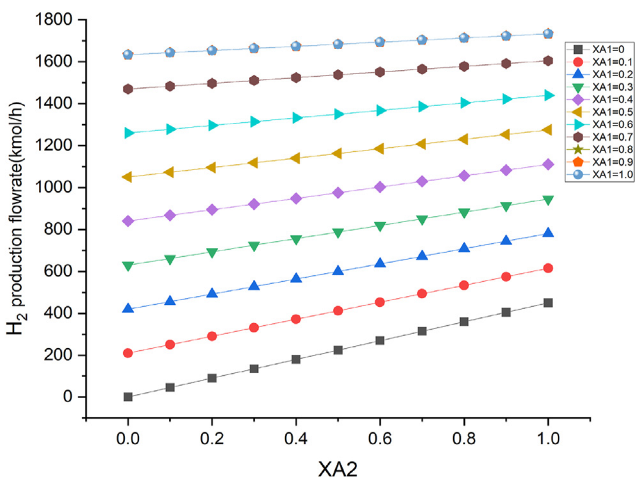

Fig. 4 – Effect of conversion of glycerol in a series reaction for Model 1.

conducted for both reactions in a series manner. The findings indicate that an increase in XA1 from 0 to 1 led to a substantial rise in H_2 production. However, when XA2 increases from 0 to 1, the increase in H_2 production is comparatively lower in independent reactions and even lower in series reactions for both R1 and R2. It is evident that under the extreme condition where XA2 is equal to zero and XA1 ranges from 0 to 1, the production of H_2 varies from 0 to 1633.3 kmol/h. Under different extreme conditions, specifically when XA2 is equal to 1 and XA1 ranges from 0 to 1, the production of H_2 exhibits a variation ranging from 450 to 1633.3 kmol/h. Fig. 4 illustrates various conversion combinations, elucidating the relationship between H_2 production and the variables XA1 and XA2.

3.5. *Comparison of parallel and series reactions*

Both series and parallel configurations are possible for reactions, depending on how the pathways are linked. The reaction kinetics and product distribution can be drastically altered by whether the reactions occur in parallel or series.

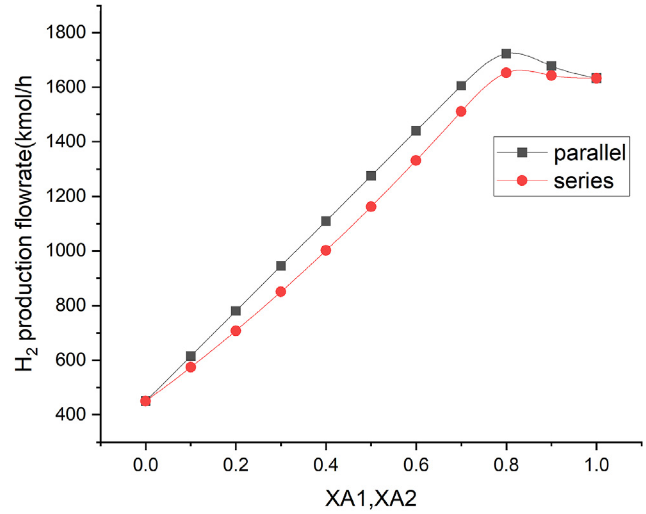

Fig. 5 – Comparison of series and parallel reactions for Model 1.

Fig. 5 analyzes the maximal H_2 generation flow rate for both parallel and series configurations of the reactions. More H_2 is generated in a parallel configuration than in a series configuration. Under both extreme conditions, the production of H_2 is equivalent for both the parallel and series configurations. According to the findings presented in Fig. 5, it can be identified that when complete conversion is achievable, any configuration can be utilized. However, in scenarios of incomplete conversion, the parallel configuration is preferred.

3.6. *Effect of temperature on RGibbs reactor*

The temperature of reactor varies from 400 to 900 °C at atmospheric pressure, and is shown in Fig. 6. It is found that as temperature increases from 400 to 658 °C, H_2 production goes on increasing and reaches its maximum at 658 °C, decreasing with further increases in temperature.

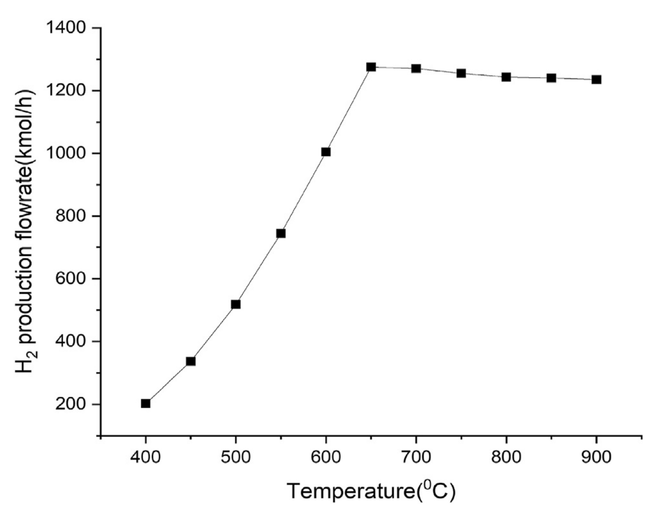

Fig. 6 – Effect of reformer temperature on H_2 production for Model 2.

Table 6 – Thermodynamically feasible results of Model 1.

XA1	XA2	H_2 flowrate (kmol/h)	Reactor temperature (^{0}C)	H_2 fraction
0	1	450	474.6	0.22
0.1	0.9	615	515.3	0.28
0.2	0.8	780	551.4	0.34
0.3	0.7	945	586.4	0.40
0.4	0.6	1110	620.6	0.45
0.5	0.5	1275	658	0.49

Table 7 – Effect of conversion on net heat duty reduction (%).

XA1	XA2	Net heat duty (kW) (Without B4)	Net heat duty (kW) (With B4)	Energy reduction (%)
0	1	46254.1	27783.6	39.9
0.1	0.9	43888.3	27081.3	38.3
0.2	0.8	41522.5	26379.0	36.5
0.3	0.7	39156.5	25676.6	34.4
0.4	0.6	36790.5	24974.3	32.1
0.5	0.5	34424.4	24271.9	29.5

3.7. *Relationship between two models*

Thermodynamic feasibility of Model 1 is done in Model 2, and results are tabulated in Table 6. Model 2 is run for a temperature range of 400–900 °C at atmospheric pressure and compares H_2 production values with Model 1. It is found that Model 1 is feasible only up to a 50% conversion of both reactions in parallel fashion. Model 1 produces 62.5 kmol/h of carbon at 50% conversion of both reactions in parallel fashion, while model 2 produces zero kmol/h of carbon at 658 °C because optimum conditions are applied.

3.8. *Effect of conversion on reactor duty*

Here, the percentage reduction of the reactor's net heat duty is calculated for both the proposed model and the literature model [56]. It is found that, if this model is used, the reactor's net heat duty will be significantly reduced because this model integrates heat. At XA1 = 0.5 and XA2 = 0.5, 29.5% less energy is consumed for maximum H_2 generation. Table 7 presents the impact of conversion on the reduction of net heat duty. The presented table provides an overview of the energy savings achieved through the implementation of heat integration techniques.

4. Conclusions

The present study employs GSR combined with Aspen Plus software to a devise H_2 production process characterized by optimal energy utilization. Two putative models may be used to conduct thermodynamic and stoichiometric evaluations. Stoichiometric principles form the basis of Model 1, while thermodynamics form the basis of model 2. The optimal molar feed ratio of water to glycerol is determined by systematically varying the quantities of water and glycerol while maintaining a feed basis of 1000 kmol/h. The optimal molar feed ratio of water to glycerol was determined to be 2.33. Subsequently, an investigation is conducted to determine the reaction pathway that yields a higher production of H_2, specifically examining the parallel and series pathways. Research has demonstrated that the parallel configuration of reactions yields a higher production of H_2 when compared to the series configuration. It is found that a H_2 production rate of 1723.3 kmol/h can be attained with a glycerol conversion rate of 0.8, accompanied by a corresponding H_2 mole fraction of 0.66. Given that Model 1 is founded upon the principles of stoichiometry, it is imperative to conduct additional research to ascertain its thermodynamic viability. At a temperature of 658 °C and a pressure of 1 atm, the RGibbs model yielded a production rate of 1275 kmol/h of H_2 with a H_2 mole fraction of 0.49. The investigation involves comparing the H_2 production values of Model 2 and Model 1 across a temperature range of 400–900 °C, while keeping atmospheric pressure constant at standard levels. The empirical observation demonstrates that a H_2 fraction of 0.49 is associated with a conversion rate of 50% for both reactions. In Model 1, a total of 62.5 kmol/h of carbon is generated through parallel reactions, with each reaction achieving a 50% conversion rate. Conversely, Model 2 does not yield any carbon production at a temperature of 658 °C. The current investigation demonstrates that the inclusion of heat integration in the model results in a noteworthy decrease in the net heat duty of the reactor. When XA1 and XA2 are both equal to 0.5, a reduction of 29.5% is observed. Future research efforts are focused on the assessment and enhancement of H_2 generation models that utilize a variety of feedstocks in a manner that is both energy-efficient and environmentally sustainable. The upcoming research efforts will focus on examining the kinetic parameters relevant to the current study along with conducting a thorough analysis of the catalyst's activity.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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